made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselv
move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

Vector space models, sparse text matrices and NMF

MATH 2740 – Mathematics of Data Science – Unit U11

Julien Arino
julien.arino@umanitoba.ca

Department of Mathematics @ University of Manitoba

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Outline

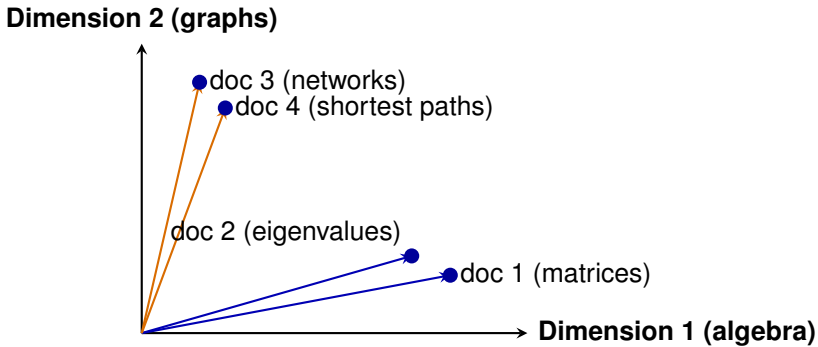
Why represent text using linear algebra?

- Much of the data generated in science, industry and daily life consists of unstructured text: books, articles, clinical notes and web pages
- Computers cannot perform calculus or geometry directly on strings of characters
- In this unit, we turn discrete text corpora into **sparse non-negative matrices** and apply three core tools of linear algebra:
  1. **Diagonal matrix scaling** (TF-IDF) to reweight rows and columns
  2. **Inner products and angles** (cosine similarity) to measure semantic similarity in high dimensions
  3. **Constrained low-rank factorisation** (Non-negative Matrix Factorisation via Alternating Least Squares) to discover latent topics

The central idea: from words to vectors

- The fundamental challenge in natural language processing (**NLP**) and information retrieval is to translate discrete symbolic data (**words** and **documents**) into a numerical format
- Why? Mathematical models are built on vector spaces, inner products and matrix factorisations rather than dictionary definitions
- The **vector space model** (**VSM**), introduced in information retrieval by Salton, Wong and Yang (1975), represents documents and words as vectors in $\mathbb{R}^m$ so that the geometry of the space reflects semantic relationships
- Documents or words with similar meaning are mapped to vectors pointing in similar directions in $\mathbb{R}^m$

A visual metaphor for the vector space model



- Vectors for documents on linear algebra point in nearly the same direction
- Vectors for documents on graph theory cluster along a different direction
- The angle between a linear algebra document and a graph theory document is large

What is a token?

- A **token** is the fundamental discrete unit of text processed by a model: it can be a word, a word stem, a sub-word or a character
- Converting a raw character string into a sequence of tokens is called **tokenisation**
- For example, after lower-casing and punctuation removal, the sentence “The cat sat on the mat” is tokenised as

$$[\texttt{"the"}, \texttt{"cat"}, \texttt{"sat"}, \texttt{"on"}, \texttt{"the"}, \texttt{"mat"}]$$

- Each unique token in our collection is assigned an integer index $i \in \{1, \ldots, m\}$ from a fixed vocabulary, yielding an integer sequence such as $[1, 2, 3, 4, 1, 5]$

Vocabulary and standard basis (one-hot) vectors

Definition 167 (Vocabulary and tokenisation)

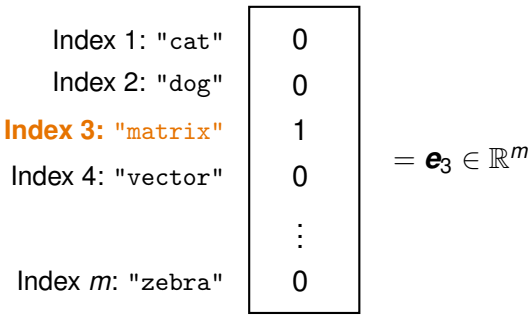
*A **vocabulary** $V = \{w_1, w_2, \ldots, w_m\}$ is an ordered set of $m = |V|$ distinct tokens. **Tokenisation** maps a raw document d to a finite sequence of tokens $(t_1, t_2, \ldots, t_L)$ with each $t_\ell \in V$*

Definition 168 (One-hot representation of a token)

*Given a vocabulary $V = \{w_1, \ldots, w_m\}$ of size m, the **one-hot vector** associated with token $w_i \in V$ is the i-th standard basis vector $\boldsymbol{e}_i \in \mathbb{R}^m$, whose k-th entry is*

$$(\boldsymbol{e}_i)_k = \delta_{ik} = \begin{cases} 1 & \text{if } k = i \\ 0 & \text{if } k \neq i \end{cases}$$

Visualising one-hot vectors and their limitations



- The set $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_m\}$ is the standard orthonormal basis of $\mathbb{R}^m$
- For any two distinct tokens $w_i \neq w_j$:

$$\boldsymbol{e}_i \bullet \boldsymbol{e}_j = 0 \quad \text{and} \quad \|\boldsymbol{e}_i - \boldsymbol{e}_j\|_2 = \sqrt{2}$$

- One-hot vectors alone treat every pair of distinct words as equally unrelated: `"matrix"` is as far from `"vector"` as it is from `"cat"`

From tokens to documents: the bag-of-words model

How do we represent a whole document d_j as a single vector in $\mathbb{R}^m$?

Definition 169 (Bag-of-words document vector)

*Let $V = \{w_1, \ldots, w_m\}$ be a vocabulary of size m and let d_j be a tokenised document $(t_1, t_2, \ldots, t_{L_j})$. The **bag-of-words** (**BoW**) vector of document d_j is*

$$\boldsymbol{x}_j = \sum_{\ell=1}^{L_j} \boldsymbol{e}_{\mathrm{idx}(t_\ell)} = \sum_{i=1}^{m} f_{ij} \boldsymbol{e}_i = \begin{pmatrix} f_{1j} \\ f_{2j} \\ \vdots \\ f_{mj} \end{pmatrix} \in \mathbb{R}^m_{\geq 0}$$

where $f_{ij} \in \mathbb{N}_0$ is the number of times term w_i occurs in document d_j

Example – Constructing bag-of-words vectors

Suppose our vocabulary has $m = 5$ terms:

$$V = \{w_1 = \text{"}\texttt{cat}\text{"},\ w_2 = \text{"}\texttt{dog}\text{"},\ w_3 = \text{"}\texttt{mat}\text{"},\ w_4 = \text{"}\texttt{sat}\text{"},\ w_5 = \text{"}\texttt{the}\text{"}\}$$

Consider two short documents (after lower-casing):

- d_1: `"the cat sat on the mat"` (ignoring `"on"` $\notin V$)
- d_2: `"the dog sat on the dog mat"`

Summing the standard basis vectors gives their bag-of-words vectors in $\mathbb{R}^5_{\geq 0}$:

$$\boldsymbol{x}_1 = 1\boldsymbol{e}_1 + 0\boldsymbol{e}_2 + 1\boldsymbol{e}_3 + 1\boldsymbol{e}_4 + 2\boldsymbol{e}_5 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 2 \end{pmatrix}, \qquad \boldsymbol{x}_2 = 0\boldsymbol{e}_1 + 2\boldsymbol{e}_2 + 1\boldsymbol{e}_3 + 1\boldsymbol{e}_4 + 2\boldsymbol{e}_5 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 1 \\ 2 \end{pmatrix}$$

What bag-of-words preserves and what it discards

- Because vector addition in $\mathbb{R}^m$ is commutative ($\boldsymbol{e}_i + \boldsymbol{e}_k = \boldsymbol{e}_k + \boldsymbol{e}_i$), the bag-of-words map completely discards **word order** and grammar
- For example, “dog bites man” and “man bites dog” produce the exact same bag-of-words vector $\boldsymbol{x} \in \mathbb{R}^m_{\geq 0}$
- Despite this simplification, bag-of-words vectors are remarkably effective for **topic classification**, **document clustering** and **search**:
  - Documents about linear algebra frequently contain `"matrix"`, `"rank"` and `"eigenvalue"`
  - Documents about graph theory frequently contain `"vertex"`, `"edge"` and `"path"`

Sparsity and non-negativity in high dimensions

Definition 170 (Support and number of non-zero entries)

For a vector $\boldsymbol{x} = (x_1, \ldots, x_m)^T \in \mathbb{R}^m$, its **support** is the index set of non-zero entries

$$\text{supp}(\boldsymbol{x}) = \{i \in \{1, \ldots, m\} : x_i \neq 0\}$$

and we write $\|\boldsymbol{x}\|_0 = |\text{supp}(\boldsymbol{x})|$ for the number of non-zero entries of $\boldsymbol{x}$

- Note that $\|\boldsymbol{x}\|_0$ is *not* a true norm (since $\|\alpha\boldsymbol{x}\|_0 = \|\boldsymbol{x}\|_0$ for $\alpha \neq 0$), but it measures **sparsity**
- In realistic text corpora, the vocabulary size m is large (10^4 to 10^5 terms), whereas a single document uses only a few hundred distinct words
- Thus $\|\boldsymbol{x}_j\|_0 \ll m$: every document vector $\boldsymbol{x}_j \in \mathbb{R}^m_{\geq 0}$ is a **high-dimensional, sparse, non-negative vector** (98% to 99.9% of its entries are zero)

Vocabulary pre-processing as row selection

In practice, before forming document vectors $\boldsymbol{x}_j \in \mathbb{R}^m_{\geq 0}$, raw tokens undergo two standard pre-processing steps:

1. **Stop-word removal**: discarding high-frequency function words (`"the"`, `"and"`, `"of"`, `"to"`) that carry little topical meaning
2. **Stemming or lemmatisation**: collapsing inflected forms (`"matrices"`, `"matrix"` $\rightarrow$ `"matrix"`; `"computing"`, `"computed"` $\rightarrow$ `"comput"`) into a single vocabulary row

- **Linear algebra view**: Removing $m - m'$ stop words from a vocabulary of size m is equivalent to left-multiplying each document vector $\boldsymbol{x}_j \in \mathbb{R}^m_{\geq 0}$ by a **row-selection matrix** $P \in \mathcal{M}_{m',m}(\{0,1\})$ obtained by deleting the stop-word rows of $\mathbb{I}_m$:

$$\boldsymbol{x}_j^{\text{reduced}} = P\boldsymbol{x}_j \in \mathbb{R}^{m'}_{\geq 0}$$

The term-document matrix

Definition 171 (Term-document matrix)

Given a vocabulary $V = \{w_1, \ldots, w_m\}$ of m terms and a corpus $D = \{d_1, \ldots, d_n\}$ of n documents, the ***term-document matrix*** *is the non-negative matrix*

$$X = [x_{ij}] = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n \end{bmatrix} \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$$

where:

- *Row $i \in \{1, \ldots, m\}$ corresponds to vocabulary term w_i*
- *Column $j \in \{1, \ldots, n\}$ is the vector $\boldsymbol{x}_j \in \mathbb{R}^m_{\geq 0}$ of document d_j*
- *Entry $x_{ij} \geq 0$ is the weight (e.g., raw frequency f_{ij}) of term w_i in document d_j*

- Some software libraries work with the transpose $X^T \in \mathcal{M}_{n,m}(\mathbb{R}_{\geq 0})$, called the **document-term matrix**; in linear algebra we prefer $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ so that documents are column vectors in $\mathbb{R}^m_{\geq 0}$

Example – A toy corpus of $n = 4$ documents and $m = 6$ terms

Consider $n = 4$ short documents across two mathematical themes, tracked over the vocabulary $V = \{\texttt{matrix}, \texttt{vector}, \texttt{eigenvalue}, \texttt{graph}, \texttt{network}, \texttt{data}\}$ ($m = 6$):

- **Document d_1 (linear algebra)**: mentions `matrix` (3 times), `vector` (2 times), `eigenvalue` (2 times) and `data` (4 times)
- **Document d_2 (linear algebra)**: mentions `matrix` (2 times), `vector` (3 times) and `data` (3 times)
- **Document d_3 (graph theory)**: mentions `graph` (3 times), `network` (2 times) and `data` (5 times)
- **Document d_4 (graph theory)**: mentions `vector` (1 time), `graph` (2 times), `network` (3 times) and `data` (4 times)

Example – Explicit 6×4 term-document matrix X

Assembling the document vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4 \in \mathbb{R}^6_{\geq 0}$ as columns gives $X \in \mathcal{M}_{6,4}(\mathbb{R}_{\geq 0})$:

Term	d_1	d_2	d_3	d_4
`matrix`	3	2	0	0
`vector`	2	3	0	1
`eigenvalue`	2	0	0	0
`graph`	0	0	3	2
`network`	0	0	2	3
`data`	4	3	5	4

$$\Longrightarrow \qquad X = \begin{pmatrix} 3 & 2 & 0 & 0 \\ 2 & 3 & 0 & 1 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 2 & 3 \\ 4 & 3 & 5 & 4 \end{pmatrix}$$

- Notice the near-block structure in rows 1–5 separating $\{d_1, d_2\}$ from $\{d_3, d_4\}$
- In contrast, row 6 (`data`) has large entries across **all four columns**, obscuring the two topics!

Dual geometric views of $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$

Every term-document matrix $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ admits two complementary geometric interpretations:

1. Column view (document space $\mathbb{R}^m_{\geq 0}$)

- Each column $\boldsymbol{x}_j \in \mathbb{R}^m_{\geq 0}$ $(j = 1, \ldots, n)$ represents a **document** as a point in m-dimensional term space
- Used to compare, cluster or classify documents

2. Row view (term space $\mathbb{R}^n_{\geq 0}$)

- Each transposed row $\boldsymbol{t}_i \in \mathbb{R}^n_{\geq 0}$ $(i = 1, \ldots, m)$ represents a **term** via its occurrence profile across the n documents
- Two words that co-occur in the same documents have similar row vectors

Why raw term counts fail

Using raw frequency counts $x_{ij} = f_{ij}$ directly in X suffers from two major defects:

1. **Common words dominate rare discriminative words**
   - Words like `"the"`, `"is"` or `"data"` appear in almost every document with large counts, dominating Euclidean norms and inner products
   - Rare domain terms like `"eigenvalue"` or `"isomorphism"` occur fewer times but carry far more information about a document's topic

2. **Document length distorts vector magnitude**
   - A 20-page article and a 1-paragraph abstract on the exact same topic have proportional term frequencies, but the longer document has a much larger norm $\|\boldsymbol{x}_j\|_2$

Term frequency and inverse document frequency (TF-IDF)

Definition 172 (Document frequency, IDF and TF-IDF)

Let $X = [f_{ij}] \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ be a raw term-document count matrix for m terms and n documents:

1. *The **term frequency** tf_{ij} is the raw count f_{ij} (or a scaled variant such as $f_{ij} / \sum_{k=1}^{m} f_{kj}$)*
2. *The **document frequency** df_i of term w_i is the number of documents in which w_i appears:*
$$\mathrm{df}_i = \left|\{j \in \{1, \ldots, n\} : f_{ij} > 0\}\right| \in \{1, \ldots, n\}$$
3. *The **inverse document frequency** of term w_i is*
$$\mathrm{idf}_i = \log\left(\frac{n}{\mathrm{df}_i}\right) \geq 0$$
4. *The **TF-IDF weight** of term w_i in document d_j is $\mathrm{tfidf}_{ij} = \mathrm{tf}_{ij} \cdot \mathrm{idf}_i$*

How $\text{idf}_i = \log(n/\text{df}_i)$ reweights vocabulary terms

Assume every vocabulary term $w_i \in V$ appears in at least one document ($1 \leq \text{df}_i \leq n$):

Term behaviour	df_i	$\text{idf}_i = \log(n/\text{df}_i)$	**Effect on row i of X**
Appears in all documents	n	$\log(1) = 0$	Zeroed out completely
Appears in half of documents	$n/2$	$\log(2) \approx 0.693$	Moderately weighted
Appears in 1 document only	1	$\log(n)$ (maximal)	Strongly amplified

- Note: throughout this course, $\log(\cdot)$ denotes the natural logarithm $\ln(\cdot)$ (changing the base of the logarithm only multiplies all idf_i by a positive constant)
- In practice, when out-of-corpus queries might have $\text{df}_i = 0$, smoothed variants such as $\log\left(\frac{n+1}{\text{df}_i+1}\right) + 1$ are also used

TF-IDF and length normalisation as diagonal matrix scaling

Theorem 173 (Row and column scaling via diagonal matrices)

Let $X = [x_{ij}] \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ be a term-document matrix with non-zero columns $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$. Define the two diagonal matrices

$$D_{IDF} = \operatorname{diag}(\mathrm{idf}_1, \ldots, \mathrm{idf}_m) \in \mathcal{M}_m(\mathbb{R}_{\geq 0}), \qquad D_{len}^{-1} = \operatorname{diag}\left(\frac{1}{\|\boldsymbol{x}_1\|}, \ldots, \frac{1}{\|\boldsymbol{x}_n\|}\right) \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$$

Then:

1. ***Left-multiplication*** $X_{TF\text{-}IDF} = D_{IDF}X$ *scales the* ***rows*** *(terms) of X: entry (i, j) is* $\mathrm{idf}_i \, x_{ij}$
2. ***Right-multiplication*** XD_{len}^{-1} *scales the* ***columns*** *(documents) of X: entry (i, j) is $x_{ij}/\|\boldsymbol{x}_j\|$*
3. *The combined matrix $\tilde{X} = D_{IDF}XD_{len}^{-1} \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ has entries $\tilde{x}_{ij} = \frac{\mathrm{idf}_i \, x_{ij}}{\|\boldsymbol{x}_j\|}$*

Proof of Theorem ??

Let $D_{\text{IDF}} = [d_{ik}] \in \mathcal{M}_m$ with $d_{ii} = \text{idf}_i$ and $d_{ik} = 0$ for $k \neq i$. By definition of matrix multiplication, the (i, j) entry of $D_{\text{IDF}}X$ is

$$(D_{\text{IDF}}X)_{ij} = \sum_{k=1}^{m} d_{ik}x_{kj} = d_{ii}x_{ij} = \text{idf}_i\, x_{ij}$$

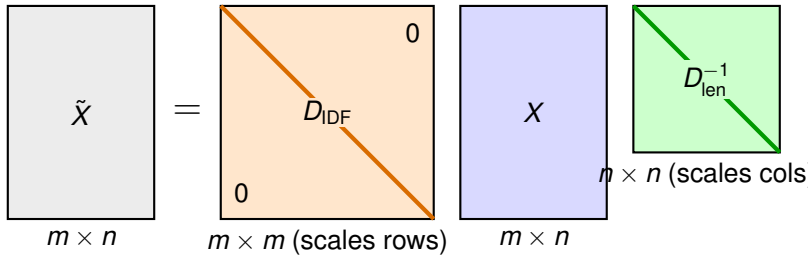
Thus row i of X is multiplied by the scalar idf_i

Similarly, let $D_{\text{len}}^{-1} = [c_{\ell j}] \in \mathcal{M}_n$ with $c_{jj} = 1/\|\boldsymbol{x}_j\|$ and $c_{\ell j} = 0$ for $\ell \neq j$. For any matrix $A = [a_{i\ell}] \in \mathcal{M}_{m,n}$, the (i, j) entry of AD_{len}^{-1} is

$$(AD_{\text{len}}^{-1})_{ij} = \sum_{\ell=1}^{n} a_{i\ell}c_{\ell j} = a_{ij}c_{jj} = \frac{a_{ij}}{\|\boldsymbol{x}_j\|}$$

Setting $A = D_{\text{IDF}}X$ gives $\tilde{x}_{ij} = (D_{\text{IDF}}XD_{\text{len}}^{-1})_{ij} = \frac{\text{idf}_i\, x_{ij}}{\|\boldsymbol{x}_j\|}$ □

Visualising $\tilde{X} = D_{\text{IDF}} X D_{\text{len}}^{-1}$



- Notice the fundamental pattern from elementary linear algebra:
  - **Left-multiplying** by a diagonal matrix scales the **rows** (the m vocabulary terms)
  - **Right-multiplying** by a diagonal matrix scales the **columns** (the n documents)

Example – Computing $D_{\text{IDF}}X$ by hand (Part 1)

Consider a corpus of $n = 4$ documents over $m = 4$ terms with raw term-document matrix $X \in \mathcal{M}_{4,4}(\mathbb{R}_{\geq 0})$:

Term	d_1	d_2	d_3	d_4
w_1 (`matrix`)	3	2	0	0
w_2 (`eigenvalue`)	2	0	0	0
w_3 (`graph`)	0	0	1	3
w_4 (`data`)	5	4	6	5

$$\Longrightarrow \qquad X = \begin{pmatrix} 3 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 5 & 4 & 6 & 5 \end{pmatrix}$$

Step 1: Count non-zero entries in each row to find df_i **and** $\text{idf}_i = \log(4/\text{df}_i)$:

- Row 1 (w_1): appears in $d_1, d_2 \implies \text{df}_1 = 2 \implies \text{idf}_1 = \log(4/2) = \log 2$
- Row 2 (w_2): appears in d_1 only
  $\implies \text{df}_2 = 1 \implies \text{idf}_2 = \log(4/1) = \log 4 = 2\log 2$
- Row 3 (w_3): appears in $d_3, d_4 \implies \text{df}_3 = 2 \implies \text{idf}_3 = \log(4/2) = \log 2$
- Row 4 (w_4): appears in
  $d_1, d_2, d_3, d_4 \implies \text{df}_4 = 4 \implies \text{idf}_4 = \log(4/4) = \log 1 = 0$

Example – Computing $D_{\text{IDF}}X$ by hand (Part 2)

Step 2: Form D_{IDF} and left-multiply X:

$$D_{\text{IDF}} = \operatorname{diag}(\log 2,\ 2\log 2,\ \log 2,\ 0) = (\log 2)\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Multiplying each row of X by its diagonal entry gives

$$X_{\text{TF-IDF}} = D_{\text{IDF}}X = (\log 2)\begin{pmatrix} 3 & 2 & 0 & 0 \\ 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} 2.08 & 1.39 & 0 & 0 \\ 2.77 & 0 & 0 & 0 \\ 0 & 0 & 0.69 & 2.08 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- Row 4 (`data`) is annihilated because it carries zero discriminative information across this corpus, whereas the rare term `eigenvalue` in Row 2 has its relative weight doubled!

Example – Column normalisation $D_{\text{IDF}} X D_{\text{len}}^{-1}$ (Part 3)

Step 3: Right-multiply $X_{\text{TF-IDF}}$ by D_{len}^{-1} to normalise the columns in ℓ_2 norm:

- The Euclidean norms of the four columns of $X_{\text{TF-IDF}} = D_{\text{IDF}} X$ are:

$$\|\boldsymbol{x}_1^{\text{TF-IDF}}\|_2 = (\log 2)\sqrt{3^2 + 4^2} = 5 \log 2, \quad \|\boldsymbol{x}_2^{\text{TF-IDF}}\|_2 = (\log 2)\sqrt{2^2 + 0^2} = 2 \log 2$$
$$\|\boldsymbol{x}_3^{\text{TF-IDF}}\|_2 = (\log 2)\sqrt{1^2} = \log 2, \quad \|\boldsymbol{x}_4^{\text{TF-IDF}}\|_2 = (\log 2)\sqrt{3^2} = 3 \log 2$$

- Right-multiplying by $D_{\text{len}}^{-1} = (\log 2)^{-1} \text{diag}(1/5,\ 1/2,\ 1,\ 1/3)$ cancels $\log 2$ and yields unit-norm columns:

$$\tilde{X} = D_{\text{IDF}} X D_{\text{len}}^{-1} = \begin{pmatrix} 3/5 & 1 & 0 & 0 \\ 4/5 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0.6 & 1 & 0 & 0 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Why Euclidean distance fails for sparse document vectors

Suppose we want to measure how similar two documents $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m_{\geq 0}$ are. Why not use the Euclidean distance $\|\boldsymbol{x} - \boldsymbol{y}\|_2$?

- Consider three documents over $V = \{\texttt{matrix}, \texttt{vector}, \texttt{graph}\}$:

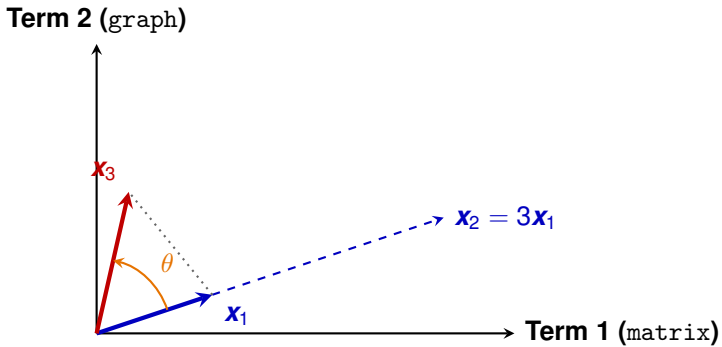
$$\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} 5 \\ 5 \\ 0 \end{pmatrix} = 5\boldsymbol{x}_1, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- Documents d_1 and d_2 have the **exact same word proportions** (pure linear algebra), whereas d_3 shares **no words at all** with d_1
- Yet their Euclidean distances are:

$$\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2 = \left\| \begin{pmatrix} -4 \\ -4 \\ 0 \end{pmatrix} \right\|_2 = \sqrt{32} \approx 5.66 \quad > \quad \|\boldsymbol{x}_1 - \boldsymbol{x}_3\|_2 = \left\| \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\|_2 = \sqrt{3} \approx 1.73$$

- Euclidean distance judges d_1 to be much closer to the unrelated short document d_3 than to a longer copy of itself!

Direction matters more than magnitude



- Scaling a document vector by a positive constant ($\boldsymbol{x}_2 = \alpha \boldsymbol{x}_1$ with $\alpha > 0$) changes its length $\|\boldsymbol{x}\|_2$ but preserves its **direction**
- Therefore, semantic similarity between documents should be measured by the **angle** θ between their vectors

Cosine similarity

Definition 174 (Cosine similarity)

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m$ be two non-zero vectors. The ***cosine similarity*** *between $\boldsymbol{x}$ and $\boldsymbol{y}$ is the cosine of the angle $\theta \in [0, \pi]$ between them:*

$$\text{sim}(\boldsymbol{x}, \boldsymbol{y}) = \cos(\theta) = \frac{\boldsymbol{x} \bullet \boldsymbol{y}}{\|\boldsymbol{x}\|_2 \, \|\boldsymbol{y}\|_2} = \frac{\sum_{i=1}^m x_i y_i}{\sqrt{\sum_{i=1}^m x_i^2} \sqrt{\sum_{i=1}^m y_i^2}}$$

Equivalently, in terms of the ℓ_2-normalised unit vectors $\hat{\boldsymbol{x}} = \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|_2}$ and $\hat{\boldsymbol{y}} = \frac{\boldsymbol{y}}{\|\boldsymbol{y}\|_2}$,

$$\text{sim}(\boldsymbol{x}, \boldsymbol{y}) = \hat{\boldsymbol{x}} \bullet \hat{\boldsymbol{y}} = \hat{\boldsymbol{x}}^T \hat{\boldsymbol{y}}$$

Fundamental properties of cosine similarity

Proposition 175 (Properties of cosine similarity)

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m$ be non-zero vectors. Then:

1. ***Positive scale invariance**: For any scalars $\alpha > 0$ and $\beta > 0$,*

$$\text{sim}(\alpha \boldsymbol{x}, \beta \boldsymbol{y}) = \text{sim}(\boldsymbol{x}, \boldsymbol{y})$$

2. ***General bounds**: By the Cauchy-Schwarz inequality, $\text{sim}(\boldsymbol{x}, \boldsymbol{y}) \in [-1, 1]$, with $\text{sim}(\boldsymbol{x}, \boldsymbol{y}) = 1 \iff \boldsymbol{y} = c\boldsymbol{x}$ for some $c > 0$*
3. ***Non-negative orthant bounds**: If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m_{\geq 0}$ are non-negative vectors, then*

$$0 \leq \text{sim}(\boldsymbol{x}, \boldsymbol{y}) \leq 1$$

Furthermore, for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m_{\geq 0}$,

$$\text{sim}(\boldsymbol{x}, \boldsymbol{y}) = 0 \iff \text{supp}(\boldsymbol{x}) \cap \text{supp}(\boldsymbol{y}) = \emptyset$$

Proof of Proposition ?? (Part 1: scale invariance)

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$ and let $\alpha, \beta > 0$. Using the bilinearity of the dot product and the positive homogeneity of the Euclidean norm ($\|\alpha \boldsymbol{x}\|_2 = |\alpha| \, \|\boldsymbol{x}\|_2 = \alpha \|\boldsymbol{x}\|_2$ since $\alpha > 0$), we compute

$$\begin{aligned}
\text{sim}(\alpha \boldsymbol{x}, \beta \boldsymbol{y}) &= \frac{(\alpha \boldsymbol{x}) \bullet (\beta \boldsymbol{y})}{\|\alpha \boldsymbol{x}\|_2 \, \|\beta \boldsymbol{y}\|_2} \\
&= \frac{\alpha\beta (\boldsymbol{x} \bullet \boldsymbol{y})}{(\alpha \|\boldsymbol{x}\|_2)(\beta \|\boldsymbol{y}\|_2)} \\
&= \frac{\cancel{\alpha\beta} (\boldsymbol{x} \bullet \boldsymbol{y})}{\cancel{\alpha\beta} \|\boldsymbol{x}\|_2 \, \|\boldsymbol{y}\|_2} \\
&= \text{sim}(\boldsymbol{x}, \boldsymbol{y})
\end{aligned}$$

(Note: if $\alpha < 0$ and $\beta > 0$, then $|\alpha| = -\alpha$, so $\text{sim}(\alpha \boldsymbol{x}, \beta \boldsymbol{y}) = -\text{sim}(\boldsymbol{x}, \boldsymbol{y})$) □

Proof of Proposition ?? (Part 2: non-negative vectors)

Assume $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m_{\geq 0} \setminus \{\mathbf{0}\}$, so $x_i \geq 0$ and $y_i \geq 0$ for every $i \in \{1, \ldots, m\}$

1. Since the product of two non-negative real numbers is non-negative, $x_i y_i \geq 0$ for all $i = 1, \ldots, m$. Summing over i yields

$$\boldsymbol{x} \bullet \boldsymbol{y} = \sum_{i=1}^{m} x_i y_i \geq 0$$

   Dividing by $\|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2 > 0$ and combining with Cauchy-Schwarz gives $0 \leq \mathrm{sim}(\boldsymbol{x}, \boldsymbol{y}) \leq 1$

2. Because each summand $x_i y_i$ is non-negative, a sum of non-negative terms $\sum_{i=1}^{m} x_i y_i$ equals 0 if and only if **every individual term is zero**:

$$\mathrm{sim}(\boldsymbol{x}, \boldsymbol{y}) = 0 \iff \sum_{i=1}^{m} x_i y_i = 0 \iff \forall i \in \{1, \ldots, m\},\ x_i y_i = 0$$

   Since $x_i y_i = 0 \iff (x_i = 0 \text{ or } y_i = 0)$, there is no index i with both $x_i > 0$ and $y_i > 0$, which means $\mathrm{supp}(\boldsymbol{x}) \cap \mathrm{supp}(\boldsymbol{y}) = \emptyset$ □

Interpretation of Proposition ?? for text

- Because bag-of-words and TF-IDF vectors lie in the non-negative orthant $\mathbb{R}^m_{\geq 0}$, two documents can **never** have negative cosine similarity ($\theta \in [0, \pi/2]$)

- $\text{sim}(\boldsymbol{x}_i, \boldsymbol{x}_j) = 0$ ($\theta = \pi/2$, orthogonal vectors) occurs **if and only if the two documents share no vocabulary words in common** (after any zero-IDF stop words are removed)

- Moreover, for ℓ_2-normalised vectors $\hat{\boldsymbol{x}} = \boldsymbol{x}/\|\boldsymbol{x}\|_2$ and $\hat{\boldsymbol{y}} = \boldsymbol{y}/\|\boldsymbol{y}\|_2$ on the unit sphere S^{m-1}, Euclidean distance and cosine similarity are monotonically related:

$$\|\hat{\boldsymbol{x}} - \hat{\boldsymbol{y}}\|_2^2 = \|\hat{\boldsymbol{x}}\|_2^2 + \|\hat{\boldsymbol{y}}\|_2^2 - 2(\hat{\boldsymbol{x}} \bullet \hat{\boldsymbol{y}}) = 2\,(1 - \text{sim}(\boldsymbol{x}, \boldsymbol{y}))$$

All-pairs cosine similarity via the Gram matrix

How do we compute the cosine similarity between **all pairs** of n documents in a matrix $X = [\boldsymbol{x}_1 \cdots \boldsymbol{x}_n] \in \mathcal{M}_{m,n}$ at once?

Theorem 176 (Cosine similarity Gram matrix)

Let $X = [\boldsymbol{x}_1 \cdots \boldsymbol{x}_n] \in \mathcal{M}_{m,n}$ *have non-zero columns and let*

$$\hat{X} = XD_{norm}^{-1} = \begin{bmatrix} \hat{\boldsymbol{x}}_1 & \hat{\boldsymbol{x}}_2 & \cdots & \hat{\boldsymbol{x}}_n \end{bmatrix} \in \mathcal{M}_{m,n} \quad where \quad D_{norm}^{-1} = \operatorname{diag}\left(\frac{1}{\|\boldsymbol{x}_1\|_2}, \ldots, \frac{1}{\|\boldsymbol{x}_n\|_2} \right)$$

Then the symmetric ***Gram matrix***

$$S = \hat{X}^T \hat{X} \in \mathcal{M}_n(\mathbb{R})$$

has (i, j) *entry equal to the pairwise cosine similarity* $s_{ij} = \hat{\boldsymbol{x}}_i^T \hat{\boldsymbol{x}}_j = \operatorname{sim}(\boldsymbol{x}_i, \boldsymbol{x}_j)$*, with* $s_{ii} = 1$ *along the diagonal*

Example – Pairwise cosine similarity by hand (Part 1)

Consider $n = 3$ sparse document vectors in $\mathbb{R}^5_{\geq 0}$:

$$\boldsymbol{x}_1 = \begin{pmatrix} 3 \\ 4 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} 6 \\ 8 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \\ 1 \end{pmatrix}$$

Step 1: Compute the Euclidean norm of each document vector:

$$\|\boldsymbol{x}_1\|_2 = \sqrt{3^2 + 4^2 + 0 + 0 + 0} = \sqrt{25} = 5$$
$$\|\boldsymbol{x}_2\|_2 = \sqrt{6^2 + 8^2 + 0 + 0 + 0} = \sqrt{100} = 10$$
$$\|\boldsymbol{x}_3\|_2 = \sqrt{0 + 2^2 + 0 + 2^2 + 1^2} = \sqrt{9} = 3$$

Example – Pairwise cosine similarity by hand (Part 2)

Step 2: Compute pairwise dot products (only shared non-zero indices contribute!):

$$\begin{aligned}
\boldsymbol{x}_1 \bullet \boldsymbol{x}_2 &= 3(6) + 4(8) = 18 + 32 = 50 \\
\boldsymbol{x}_1 \bullet \boldsymbol{x}_3 &= 4(2) = 8 \\
\boldsymbol{x}_2 \bullet \boldsymbol{x}_3 &= 8(2) = 16
\end{aligned}$$

Step 3: Divide by norms to form the Gram matrix $S = \hat{X}^T \hat{X}$:

$$\text{sim}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{50}{5 \times 10} = 1, \quad \text{sim}(\boldsymbol{x}_1, \boldsymbol{x}_3) = \frac{8}{5 \times 3} = \frac{8}{15}, \quad \text{sim}(\boldsymbol{x}_2, \boldsymbol{x}_3) = \frac{16}{10 \times 3} = \frac{8}{15}$$

$$\implies S = \hat{X}^T \hat{X} = \begin{pmatrix} 1 & 1 & 8/15 \\ 1 & 1 & 8/15 \\ 8/15 & 8/15 & 1 \end{pmatrix} \approx \begin{pmatrix} 1.000 & 1.000 & 0.533 \\ 1.000 & 1.000 & 0.533 \\ 0.533 & 0.533 & 1.000 \end{pmatrix}$$

Recall from Unit 04: low-rank approximation via SVD

In Unit 04, we saw that any matrix $X \in \mathcal{M}_{m,n}(\mathbb{R})$ admits a Singular Value Decomposition $X = U\Sigma V^T = \sum_{r=1}^{\text{rank}(X)} \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^T$

- By the **Eckart-Young theorem**, the best rank-k approximation to X in the Frobenius norm $\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}$ is the **truncated SVD**:

$$X_k = \sum_{r=1}^{k} \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^T = U_k \Sigma_k V_k^T$$

- Applied to a term-document matrix X, truncated SVD is known in information retrieval as **Latent Semantic Analysis** (**LSA**, Deerwester et al. 1990)

Why truncated SVD is awkward on non-negative data

Many real-world data matrices are inherently **non-negative** ($x_{ij} \geq 0$):

- Term-document word count and TF-IDF matrices
- Pixel intensities in greyscale or colour images
- Audio spectrograms, chemical concentrations and gene expression counts

Why does truncated SVD struggle with interpretability on $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$?

- By the Perron-Frobenius theorem, the first singular vectors $\boldsymbol{u}_1, \boldsymbol{v}_1$ can be chosen non-negative
- However, U_k and V_k have **orthonormal columns**: for every $r \geq 2$,

$$\boldsymbol{u}_r \bullet \boldsymbol{u}_1 = 0 \quad \text{and} \quad \boldsymbol{v}_r \bullet \boldsymbol{v}_1 = 0$$

- As proved in Proposition **??**, a non-zero vector orthogonal to a strictly positive vector $\boldsymbol{u}_1$ **must contain both positive and negative entries**!

Subtractive cancellation vs additive parts

- Because $\boldsymbol{u}_2, \ldots, \boldsymbol{u}_k$ and $\boldsymbol{v}_2, \ldots, \boldsymbol{v}_k$ contain negative numbers, SVD reconstructs a document $\boldsymbol{x}_j \approx \sum_{r=1}^{k} (\sigma_r v_{jr}) \boldsymbol{u}_r$ through **subtractive cancellation**:
  - What does it mean for a "topic" vector $\boldsymbol{u}_2$ to contain $-0.42 \times$ "`matrix`"?
  - What does it mean to subtract 0.3 of Topic 2 from a document so that negative word counts cancel out?
- For example, on our 6×4 corpus matrix X, the first two left singular vectors in U_2 are

$$\boldsymbol{u}_1 = \begin{pmatrix} 0.30 \\ 0.30 \\ 0.11 \\ 0.32 \\ 0.31 \\ 0.78 \end{pmatrix} \geq \mathbf{0}, \qquad \boldsymbol{u}_2 = \begin{pmatrix} \textbf{-0.54} \\ \textbf{-0.47} \\ \textbf{-0.24} \\ 0.45 \\ 0.45 \\ 0.14 \end{pmatrix} \notin \mathbb{R}^6_{\geq 0}$$

- To obtain directly interpretable topics (or parts of an image), we drop the orthogonality constraint and instead enforce **non-negativity** on both factors!

Non-negative Matrix Factorisation (NMF)

Definition 177 (Non-negative Matrix Factorisation (NMF))

Given a non-negative matrix $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ and a target rank $k < \min(m, n)$, ***Non-negative Matrix Factorisation*** *(**NMF**, Paatero & Tapper 1994; Lee & Seung 1999) seeks two non-negative matrices*

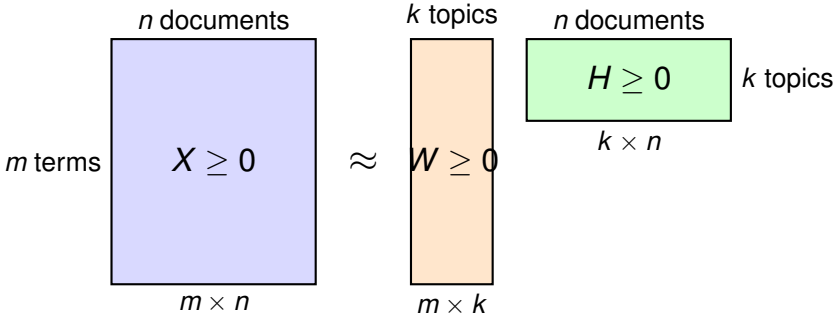
$$W \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0}) \quad \textit{and} \quad H \in \mathcal{M}_{k,n}(\mathbb{R}_{\geq 0})$$

such that $X \approx WH$, by solving the constrained optimisation problem

$$\min_{W \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0}),\, H \in \mathcal{M}_{k,n}(\mathbb{R}_{\geq 0})} \|X - WH\|_F^2 = \sum_{i=1}^{m} \sum_{j=1}^{n} \left(x_{ij} - \sum_{r=1}^{k} w_{ir} h_{rj} \right)^2$$

subject to $w_{ir} \geq 0$ and $h_{rj} \geq 0$ for all i, r, j

Visualising the low-rank NMF decomposition $X \approx WH$



- When $k \ll \min(m, n)$, NMF compresses mn entries into $k(m + n)$ non-negative parameters while preserving an **additive, parts-based** structure

Comparison: truncated SVD vs NMF

Property	**Truncated SVD ($U_k \Sigma_k V_k^T$)**	**NMF (WH)**
Structural constraint	Orthogonality ($U_k^T U_k = V_k^T V_k = \mathbb{I}_k$)	Non-negativity ($W \geq$
Entries of factors	Positive and negative ($\in \mathbb{R}$)	Non-negative only ($\in$
Reconstruction	Subtractive cancellation	Purely additive parts
Frobenius error $\|X - \hat{X}\|_F$	Globally minimal for rank k	Higher or equal ($\geq \|$
Optimisation problem	Exact via eigendecomposition	Non-convex; iterative
Nested factors across k	Yes (first k cols stay fixed)	No (recomputed for e

Column-wise view: documents as additive mixtures of topics

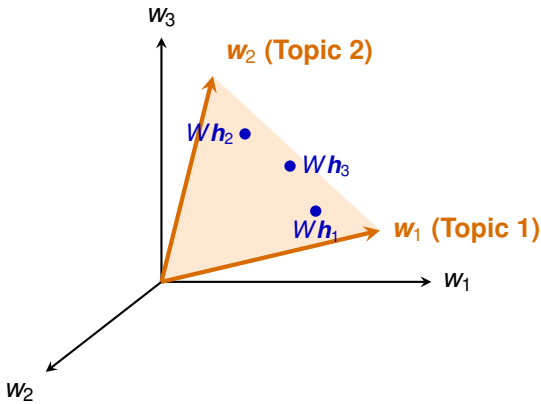
Write $W = \begin{bmatrix} \boldsymbol{w}_1 & \cdots & \boldsymbol{w}_k \end{bmatrix} \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0})$ in terms of its columns $\boldsymbol{w}_r \in \mathbb{R}^m_{\geq 0}$ and $H = \begin{bmatrix} \boldsymbol{h}_1 & \cdots & \boldsymbol{h}_n \end{bmatrix} \in \mathcal{M}_{k,n}(\mathbb{R}_{\geq 0})$ in terms of its columns $\boldsymbol{h}_j \in \mathbb{R}^k_{\geq 0}$

By standard column-wise matrix multiplication, the j-th column of $X \approx WH$ is

$$\boldsymbol{x}_j \approx W\boldsymbol{h}_j = h_{1j}\boldsymbol{w}_1 + h_{2j}\boldsymbol{w}_2 + \cdots + h_{kj}\boldsymbol{w}_k = \sum_{r=1}^{k} h_{rj}\boldsymbol{w}_r$$

- **Columns of W ($\boldsymbol{w}_r \in \mathbb{R}^m_{\geq 0}$, the dictionary / topic basis)**:
  - Column $\boldsymbol{w}_r$ gives the weight of each of the m vocabulary words in **topic** r
  - The largest entries of $\boldsymbol{w}_r$ identify the signature keywords of topic r
- **Columns of H ($\boldsymbol{h}_j \in \mathbb{R}^k_{\geq 0}$, the document topic encodings)**:
  - Entry $h_{rj} \geq 0$ tells how strongly document d_j expresses topic r

Geometric interpretation: the conical hull of W



- In unconstrained linear algebra, $\{W\boldsymbol{h} : \boldsymbol{h} \in \mathbb{R}^k\}$ is the k-dimensional subspace span$(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k)$
- In NMF, because $\boldsymbol{h}_j \in \mathbb{R}^k_{\geq 0}$ has non-negative weights $h_{rj} \geq 0$, the reconstructed documents $W\boldsymbol{h}_j$ must lie inside the **conical hull** (or simplicial cone) generated by the columns of W:

$$\text{cone}(W) = \left\{ \sum_{r=1}^{k} h_r \boldsymbol{w}_r : h_1, \ldots, h_k \geq 0 \right\} \subset$$

Rank-one outer product expansion and polysemy

Just as the SVD writes $X_k = \sum_{r=1}^{k} \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^T$ as a sum of k rank-one matrices, NMF decomposes WH into k **non-negative rank-one layers**:

$$WH = \sum_{r=1}^{k} \boldsymbol{w}_r \tilde{\boldsymbol{h}}_r^T$$

where $\boldsymbol{w}_r \in \mathbb{R}_{\geq 0}^m$ is the r-th column of W and $\tilde{\boldsymbol{h}}_r^T \in \mathbb{R}_{\geq 0}^{1 \times n}$ is the r-th row of H

- Each outer product $M_r = \boldsymbol{w}_r \tilde{\boldsymbol{h}}_r^T \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ displays the contribution of **topic** r **alone** across all m words and all n documents
- Because columns of W do not have to be orthogonal, a single word w_i with multiple meanings (**polysemy**, e.g., `"degree"` in polynomial algebra vs vertex degree in graphs) can have positive weight $w_{ir} > 0$ in **more than one topic column**!

Example – Interpreting W and H (Part 1)

Suppose a corpus of $n = 4$ documents over $m = 5$ terms $\{\texttt{matrix}, \texttt{vector}, \texttt{rank}, \texttt{graph}, \texttt{vertex}\}$ has term-document matrix $X \in \mathcal{M}_{5,4}(\mathbb{R}_{\geq 0})$ and rank-2 NMF gives $X \approx WH$:

$$X = \begin{pmatrix} 6 & 4 & 0 & 2 \\ 3 & 2 & 0 & 1 \\ 3 & 2 & 0 & 1 \\ 0 & 0 & 6 & 4 \\ 0 & 0 & 3 & 2 \end{pmatrix}, \qquad W = \begin{pmatrix} 2 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 3 & 2 & 0 & 1 \\ 0 & 0 & 3 & 2 \end{pmatrix}$$

(where rows $1, \ldots, 5$ of X and W are $\texttt{matrix}, \texttt{vector}, \texttt{rank}, \texttt{graph}, \texttt{vertex}$ while columns $1, \ldots, 4$ of X and H are d_1, d_2, d_3, d_4)

Questions:

1. How should we label Topic 1 ($\boldsymbol{w}_1$) and Topic 2 ($\boldsymbol{w}_2$) from W?
2. How should we characterise documents d_1, d_2, d_3, d_4 from H?
3. What is the reconstruction error $\|X - WH\|_F^2$?

Example – Interpreting W and H (Part 2)

1. **Labelling topics from the columns of W:**
   - Column $\boldsymbol{w}_1 = (2, 1, 1, 0, 0)^T$ has non-zero weights only on `matrix`, `vector` and `rank` $\implies$ **Topic 1 is Linear Algebra**
   - Column $\boldsymbol{w}_2 = (0, 0, 0, 2, 1)^T$ has non-zero weights only on `graph` and `vertex` $\implies$ **Topic 2 is Graph Theory**

2. **Classifying documents from the columns of H:**
   - $\boldsymbol{h}_1 = (3, 0)^T$ and $\boldsymbol{h}_2 = (2, 0)^T$ load purely on Topic 1 (Linear Algebra)
   - $\boldsymbol{h}_3 = (0, 3)^T$ loads purely on Topic 2 (Graph Theory)
   - $\boldsymbol{h}_4 = (1, 2)^T$ is a **multi-topic document**: $\boldsymbol{x}_4 = 1\boldsymbol{w}_1 + 2\boldsymbol{w}_2$ ($1/3$ Linear Algebra, $2/3$ Graph Theory, e.g., spectral graph theory!)

3. **Reconstruction error**: Multiplying WH reproduces X exactly, so $\|X - WH\|_F^2 = 0$ and $\text{rank}(X) = 2$

Why computing NMF is an optimisation challenge

Consider the NMF objective function

$$f(W,H) = \|X - WH\|_F^2 = \sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \sum_{r=1}^{k} w_{ir}h_{rj}\right)^2$$

1. **Non-uniqueness (scaling and permutation ambiguity)**:
   - For any permutation matrix $P \in \mathcal{M}_k$ and any positive diagonal matrix $D = \text{diag}(d_1, \ldots, d_k)$ with $d_r > 0$, setting $\tilde{W} = WPD \geq 0$ and $\tilde{H} = D^{-1}P^T H \geq 0$ gives $\tilde{W}\tilde{H} = WH$

2. **Non-convexity jointly in** (W, H):
   - Because unknown parameters w_{ir} and h_{rj} are multiplied together in WH, $f(W,H)$ is **not convex** in (W,H) simultaneously and can have multiple local minima

The key linear algebra insight: bi-convexity

Although $f(W, H) = \|X - WH\|_F^2$ is non-convex jointly in (W, H), it is **bi-convex** (convex in H when W is fixed and convex in W when H is fixed)!

1. **Hold** $W \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0})$ **fixed**:

$$\|X - WH\|_F^2 = \sum_{j=1}^{n} \|\boldsymbol{x}_j - W\boldsymbol{h}_j\|_2^2$$

   Each term $\|\boldsymbol{x}_j - W\boldsymbol{h}_j\|_2^2$ is a standard **linear least-squares problem** $W\boldsymbol{h}_j \approx \boldsymbol{x}_j$ of the exact form studied in Unit 03 and Unit 04!

2. **Hold** $H \in \mathcal{M}_{k,n}(\mathbb{R}_{\geq 0})$ **fixed**: Using $\|A\|_F^2 = \|A^T\|_F^2$ and $(WH)^T = H^T W^T$,

$$\|X - WH\|_F^2 = \|X^T - H^T W^T\|_F^2$$

   This is again a linear least-squares problem for the columns of W^T with design matrix $H^T \in \mathcal{M}_{n,k}$!

Deriving the projected least-squares update for H

Proposition 178 (Projected least-squares step for H)

Suppose $W \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0})$ is fixed and has full column rank ($\operatorname{rank}(W) = k$*, so $W^T W \in \mathcal{M}_k$ is invertible). Then:*

1. *For each column $j \in \{1, \ldots, n\}$, the unconstrained least-squares problem $\min_{\boldsymbol{h}_j \in \mathbb{R}^k} \|\boldsymbol{x}_j - W\boldsymbol{h}_j\|_2^2$ has the unique normal-equations solution*

$$\boldsymbol{h}_j^{LS} = (W^T W)^{-1} W^T \boldsymbol{x}_j$$

2. *Stacking all n columns side by side into $H_{LS} = [\boldsymbol{h}_1^{LS} \cdots \boldsymbol{h}_n^{LS}] \in \mathcal{M}_{k,n}(\mathbb{R})$ gives the matrix formula*

$$H_{LS} = (W^T W)^{-1} W^T X$$

3. *In projected **Alternating Least Squares** (**ALS**), we enforce non-negativity by clipping negative entries to zero entrywise:*

$$H \leftarrow \max(0, H_{LS}) \quad i.e., \quad h_{rj} = \max\big(0, (H_{LS})_{rj}\big)$$

Deriving the projected least-squares update for W

Proposition 179 (Projected least-squares step for W)

Suppose $H \in \mathcal{M}_{k,n}(\mathbb{R}_{\geq 0})$ is fixed and has full row rank ($\operatorname{rank}(H) = k$, so $HH^T \in \mathcal{M}_k$ is invertible). Then:

1. *Minimising $\|X - WH\|_F^2 = \|X^T - H^T W^T\|_F^2$ without sign constraints yields the normal equations for W^T:*

$$(H^T)^T (H^T) W_{LS}^T = (H^T)^T X^T \iff (HH^T) W_{LS}^T = HX^T$$

2. *Solving for W_{LS}^T and taking the transpose (using $((HH^T)^{-1})^T = (HH^T)^{-1}$) gives*

$$W_{LS}^T = (HH^T)^{-1} HX^T \iff W_{LS} = XH^T (HH^T)^{-1}$$

3. *Clipping negative entries to zero entrywise gives the projected ALS update:*

$$W \leftarrow \max(0, W_{LS})$$

The Alternating Least Squares (ALS) algorithm for NMF

Putting Propositions **??** and **??** together yields the **Projected Alternating Least Squares** (**ALS**) algorithm (Paatero & Tapper 1994; Berry et al. 2007):

1. **Input**: Non-negative matrix $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ and target rank $k < \min(m, n)$
2. **Initialise**: Choose a non-negative initial matrix $W^{(0)} \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0})$ with $\text{rank}(W^{(0)}) = k$ (e.g., using positive random numbers)
3. **Repeat for** $t = 0, 1, 2, \ldots$ **until convergence**:
   - 3.1 **Update** H **(fixing** $W^{(t)}$**)**: Solve $(W^{(t)T} W^{(t)}) H_{\text{LS}} = W^{(t)T} X$ and project:

$$H^{(t+1)} = \max\Big(0,\ (W^{(t)T} W^{(t)})^{-1} W^{(t)T} X\Big)$$

   - 3.2 **Update** W **(fixing** $H^{(t+1)}$**)**: Solve $W_{\text{LS}}(H^{(t+1)} H^{(t+1)T}) = X H^{(t+1)T}$ and project:

$$W^{(t+1)} = \max\Big(0,\ X H^{(t+1)T} (H^{(t+1)} H^{(t+1)T})^{-1}\Big)$$

   - 3.3 *(Optional)* Normalise each column of $W^{(t+1)}$ to unit ℓ_2 or ℓ_1 norm to prevent scaling drift

Example – One ALS update step by hand (Part 1)

Let $X \in \mathcal{M}_{3,3}(\mathbb{R}_{\geq 0})$ and let the initial rank-2 dictionary matrix $W^{(0)} \in \mathcal{M}_{3,2}(\mathbb{R}_{\geq 0})$ be

$$X = \begin{pmatrix} 3 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 1 \end{pmatrix}, \qquad W^{(0)} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Let us compute the first ALS update $H^{(1)} = \max\big(0, (W^{(0)T} W^{(0)})^{-1} W^{(0)T} X\big)$ by hand

Step 1: Compute $W^{(0)T} W^{(0)} \in \mathcal{M}_2$ and its inverse:

$$W^{(0)T} W^{(0)} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Since $\det(W^{(0)T} W^{(0)}) = 2(2) - 1(1) = 3 \neq 0$, the 2×2 inverse formula gives

$$(W^{(0)T} W^{(0)})^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Example – One ALS update step by hand (Part 2)

Step 2: Compute $W^{(0)T}X \in \mathcal{M}_{2,3}$:

$$W^{(0)T}X = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 3 & 5 \\ 0 & 6 & 4 \end{pmatrix}$$

Step 3: Compute the unconstrained least-squares matrix $H_{\mathbf{LS}}^{(1)}$:

$$H_{\text{LS}}^{(1)} = (W^{(0)T}W^{(0)})^{-1}W^{(0)T}X = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 3 & 5 \\ 0 & 6 & 4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 6 & 0 & 6 \\ -3 & 9 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ \mathbf{-1} & 3 \end{pmatrix}$$

Step 4: Project onto $\mathbb{R}_{\geq 0}$ by clipping negative entries to zero:

$$H^{(1)} = \max\left(0, H_{\text{LS}}^{(1)}\right) = \begin{pmatrix} 2 & 0 & 2 \\ \mathbf{0} & 3 & 1 \end{pmatrix} \in \mathcal{M}_{2,3}(\mathbb{R}_{\geq 0})$$

Notice how unconstrained least squares produced -1 in entry $(2,1)$, which the projection step clips to 0!

Example – Evaluating the reconstruction $W^{(0)}H^{(1)}$ (Part 3)

Step 5: Compute the rank-2 approximation $\hat{X}^{(1)} = W^{(0)}H^{(1)}$ and its Frobenius error:

$$\hat{X}^{(1)} = W^{(0)}H^{(1)} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 2 \\ 0 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 2 & 3 & 3 \\ 0 & 3 & 1 \end{pmatrix}$$

Comparing $\hat{X}^{(1)}$ with $X = \begin{pmatrix} 3 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 1 \end{pmatrix}$, the residual matrix is

$$X - W^{(0)}H^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies \|X - W^{(0)}H^{(1)}\|_F^2 = 1^2 + (-2)^2 = 5$$

- Columns 2 and 3 of X are reconstructed **with zero error** on the very first half-step; the next half-step updates $W^{(1)} = \max\big(0, XH^{(1)T}(H^{(1)}H^{(1)T})^{-1}\big)$ to reduce the error on column 1 even further!

Remark: exact NNLS and multiplicative updates

- **Projected ALS vs exact Non-Negative Least Squares (NNLS)**:
  - Solving unconstrained least squares and clipping $\max(0, \cdot)$ is fast (requiring only a $k \times k$ linear system $W^T W$ per iteration) and works well when k is small and X is sparse
  - A more accurate variant solves each column subproblem $\min_{\boldsymbol{h}_j \geq \mathbf{0}} \|\boldsymbol{x}_j - W\boldsymbol{h}_j\|_2^2$ using an exact active-set **NNLS** solver (Lawson & Hanson 1974), guaranteeing that $\|X - W^{(t)}H^{(t)}\|_F^2$ decreases monotonically at every step

- **Lee & Seung (1999) multiplicative updates**:
  - Another popular algorithm updates entries by elementwise multiplication:

$$h_{rj} \leftarrow h_{rj} \frac{(W^T X)_{rj}}{(W^T W H)_{rj}}, \qquad w_{ir} \leftarrow w_{ir} \frac{(XH^T)_{ir}}{(WHH^T)_{ir}}$$

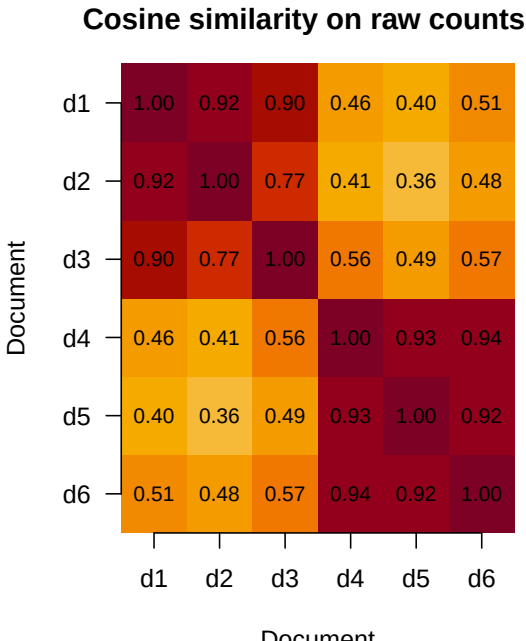
  - Because all factors on the right-hand side are non-negative, W and H automatically remain non-negative without clipping while $\|X - WH\|_F^2$ is guaranteed not to increase

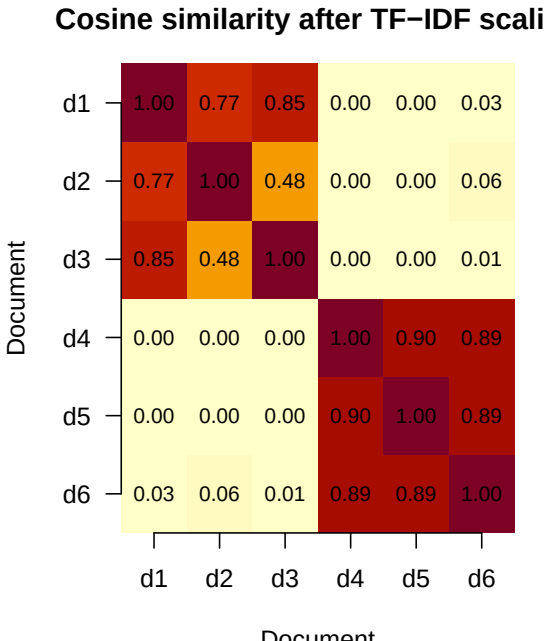
R demonstration – TF-IDF and cosine similarity

Let us build an 8×6 term-document matrix in R with 6 documents (3 on linear algebra, 3 on graphs) and compute $\tilde{X} = D_{\text{IDF}} X D_{\text{len}}^{-1}$ and $S = \hat{X}^T \hat{X}$:

```
terms = c("matrix", "vector", "eigen", "rank", "graph", "vertex", "edge", "c
X = matrix(c(
  4, 3, 2, 1, 0, 0, 0, 5,   # d1: linear algebra
  3, 4, 0, 2, 0, 0, 0, 4,   # d2: linear algebra
  2, 1, 3, 2, 0, 0, 0, 6,   # d3: linear algebra
  0, 0, 0, 0, 4, 3, 2, 5,   # d4: graph theory
  0, 0, 0, 0, 3, 2, 4, 4,   # d5: graph theory
  0, 1, 0, 0, 2, 4, 3, 5    # d6: graph theory (mentions "vector" once)
), nrow = 8, ncol = 6, dimnames = list(terms, paste0("d", 1:6)))

# Diagonal IDF scaling: D_IDF %*% X
df_i = rowSums(X > 0)
idf_i = log(ncol(X) / df_i)
X_tfidf = diag(idf_i) %*% X
```


Cosine similarity on raw counts X
d1 1.00 0.92 0.90 0.46 0.40 0.51
d2 0.92 1.00 0.77 0.41 0.36 0.48
d3 0.90 0.77 1.00 0.56 0.49 0.57
d4 0.46 0.41 0.56 1.00 0.93 0.94
d5 0.40 0.36 0.49 0.93 1.00 0.92
d6 0.51 0.48 0.57 0.94 0.92 1.00
Document
d1 d2 d3 d4 d5 d6
Document


Cosine similarity after TF−IDF scaling
d1 1.00 0.77 0.85 0.00 0.00 0.03
d2 0.77 1.00 0.48 0.00 0.00 0.06
d3 0.85 0.48 1.00 0.00 0.00 0.01
d4 0.00 0.00 0.00 1.00 0.90 0.89
d5 0.00 0.00 0.00 0.90 1.00 0.89
d6 0.03 0.06 0.01 0.89 0.89 1.00
Document
d1 d2 d3 d4 d5 d6
Document

R demonstration – Projected ALS for NMF

We now run 30 iterations of projected Alternating Least Squares with rank $k = 2$ on our TF-IDF matrix:

```
set.seed(2740)
k = 2
W = matrix(runif(nrow(X_tfidf) * k, 0.1, 1), nrow = nrow(X_tfidf), ncol = k)
H = matrix(0, nrow = k, ncol = ncol(X_tfidf))
err_history = numeric(30)

for (iter in 1:30) {
  # Update H by least squares and clip negatives to 0
  H = pmax(0, solve(t(W) %*% W + 1e-8 * diag(k), t(W) %*% X_tfidf))
  # Update W by least squares and clip negatives to 0
  W = pmax(0, t(solve(H %*% t(H) + 1e-8 * diag(k), H %*% t(X_tfidf))))
  # Normalise columns of W and rescale rows of H accordingly
  w_norms = sqrt(colSums(W^2))
  W = W %*% diag(1 / w_norms)
```

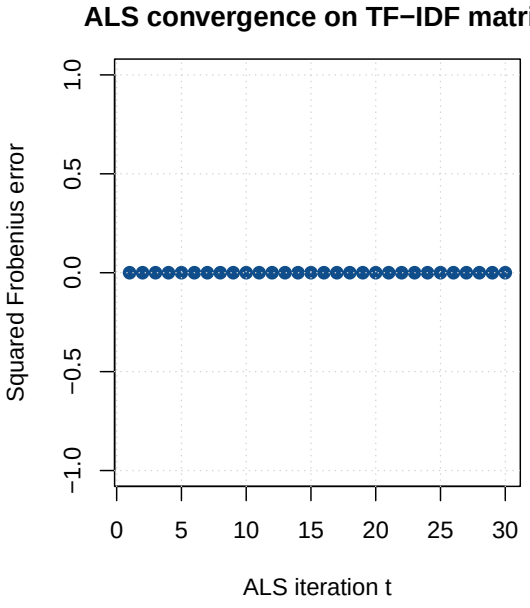
R demonstration – Inspecting the learned W and H factors

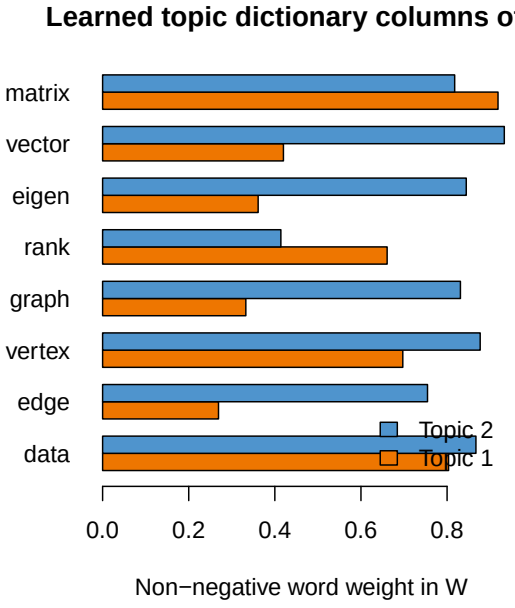
```
round(W, 3)
```

```
##        Topic_1 Topic_2
## matrix   0.918   0.818
## vector   0.420   0.933
## eigen    0.361   0.845
## rank     0.661   0.414
## graph    0.333   0.831
## vertex   0.697   0.877
## edge     0.269   0.755
## data     0.803   0.867
```

```
round(H, 2)
```

```
##  [1] 1.06 0.35 1.78 0.00 0.50 0.60 0.00 1.93 0.00 2.21 0.00 1.82
```


ALS convergence on TF−IDF matrix
Squared Frobenius error
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
30
ALS iteration t


Learned topic dictionary columns of W
matrix
vector
eigen
rank
graph
vertex
edge
data
Topic 2
Topic 1
0.0
0.2
0.4
0.6
0.8
Non−negative word weight in W

Summary of Unit U11

- **Tokenisation and bag-of-words**: A vocabulary $V = \{w_1, \ldots, w_m\}$ maps each token to $\boldsymbol{e}_i \in \mathbb{R}^m$ and each document d_j to a sparse count vector $\boldsymbol{x}_j = \sum_{i=1}^{m} f_{ij}\boldsymbol{e}_i \in \mathbb{R}^m_{\geq 0}$ with $\|\boldsymbol{x}_j\|_0 \ll m$
- **Term-document matrices and TF-IDF scaling**: Stacking document vectors as columns forms $X \in \mathcal{M}_{m,n}(\mathbb{R}_{\geq 0})$ where TF-IDF with length normalisation is the two-sided diagonal matrix product

$$\tilde{X} = D_{\text{IDF}} X D_{\text{len}}^{-1}$$

- **Cosine similarity in high dimensions**: Semantic similarity is measured by the scale-invariant angle $\text{sim}(\boldsymbol{x}, \boldsymbol{y}) = \frac{\boldsymbol{x} \bullet \boldsymbol{y}}{\|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2} \in [0, 1]$ on $\mathbb{R}^m_{\geq 0}$ whose pairwise values form the Gram matrix $S = \hat{X}^T \hat{X}$
- **Non-negative Matrix Factorisation (NMF)**: Approximating $X \approx WH$ with $W \in \mathcal{M}_{m,k}(\mathbb{R}_{\geq 0})$ and $H \in \mathcal{M}_{k,n}(\mathbb{R}_{\geq 0})$ avoids the subtractive cancellation of SVD, yielding additive topic columns $\boldsymbol{w}_r$ and document encodings $\boldsymbol{h}_j$ computed via projected **Alternating Least Squares (ALS)**